Fabrication, Not Flattery: A Taxonomy and Timeline of Reward Hacking in a Small RLHF Pipeline

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Abstract

Reinforcement learning from human feedback (RLHF) optimizes a language model against a learned reward model (RM), and the policy can raise its reward by exploiting features that the RM over-values. These behaviors have been documented separately, but rarely measured on the same training runs with per-category timing. We run a small end-to-end RLHF pipeline: an RM trained on 20,000 human pairwise votes, a 0.5B-parameter policy optimized against it for 160 updates over three seeds, and 11 checkpoints per run. We score all 10,560 checkpoint responses with the RM, with judge-free surface metrics, and with two blinded LLM judges that stand in for human raters and apply a six-category taxonomy. Most of our four pre-specified hypotheses fail. Verbosity and sycophancy together are the primary label on 49.9% of reward-hacking instances (95% CI 45.3–54.3), below the hypothesized 60%, and the verbosity share is no higher than its 47.1% base rate before training. Verbosity appears by 20% of training but plateaus after 50–60%; sycophancy never emerges; markdown structure changes first, at 10%, rather than last. The RM score and judge preference detectably diverge at 40% of training, with no earlier phase in which judges prefer the trained policy. The failure mode most enriched among reward-hacking instances is confident fabrication: 34.5% of instances versus 11.5% of untrained responses. Our judges agree on overall quality (Spearman 0.71) but poorly on most taxonomy labels (κ 0.10–0.23 for five of six), so taxonomy shares from LLM judges need human validation before they are trusted.

1 Introduction

Reinforcement learning from human feedback (RLHF) does not optimize what people want; it optimizes a learned reward model (RM) that approximates what people chose in a finite set of comparisons [Christiano et al., 2017, Stiennon et al., 2020, Ouyang et al., 2022]. When the policy finds features that the RM over-values, measured reward keeps rising while real quality stalls or falls. This is reward hacking [Amodei et al., 2016, Skalse et al., 2022], and it is a practical problem: a practitioner watching the reward curve cannot tell improvement from exploitation.

Knowing *which* proxy-exploiting behaviors dominate, and *when* they appear, tells practitioners what to monitor and when to stop. Prior work documents the individual phenomena. Reward models and RLHF-trained policies favor longer responses [Singhal et al., 2023]. Models trained on human feedback tend to agree with users' stated views [Perez et al., 2022, Sharma et al., 2024]. Optimizing far enough against a learned reward lowers true reward [Gao et al., 2023, Moskovitz et al., 2024]. However, these results come from different models, reward models, and measurement protocols. What is less common is a single taxonomy applied to the *same* training runs, so that the shares of different failure modes are comparable and each one has an onset time on a shared training axis.

Which failure modes make up reward hacking, when do they appear, and when does the RM score stop tracking human preference? We study this question with a small but complete RLHF pipeline. We train an RM on 20,000 human pairwise votes from Chatbot Arena [Chiang et al.,

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2024], optimize a QWEN2.5-0.5B-INSTRUCT policy against it for 160 updates (three seeds plus one strong-KL ablation), and save 11 checkpoints per run. We score all 10,560 checkpoint responses with the RM, with judge-free surface metrics, and with two blinded LLM judges from different model families that rate quality, compare each response with the untrained policy's response, and assign labels from a six-category taxonomy: verbosity, sycophancy, format gaming, repetition, evasion, and fabrication. No human raters were available, so the judges are a proxy for human preference throughout.

We fix four hypotheses and their operational definitions before seeing any judge labels: (H1) verbosity and sycophancy are the primary category in more than 60% of reward-hacking instances; (H2) both emerge within the first 20% of training and then rise monotonically; (H3) format gaming appears later and saturates; (H4) human preference diverges from the RM score after 40% of training.

Most of these fail. Verbosity plus sycophancy accounts for 49.9% of reward-hacking instances (95% CI 45.3–54.3) under the primary judge, and nearly all of that is verbosity: sycophancy is 0.6%. The verbosity share is not even an enrichment, because the same judge assigns verbosity as the primary label to 47.1% of responses from the *untrained* policy. Verbosity does appear by 20% of training (+22 tokens per response), but it plateaus after 50–60% rather than rising monotonically, and sycophancy does not emerge on any of three measures. Markdown structure is the earliest change, significant at 10% of training, and it peaks at 50% and then declines. The RM score and judge preference are first detectably divergent at the 40% checkpoint, which roughly matches H4 in timing, but point estimates fall from 20–30% and judges never clearly prefer the trained policy. The failure mode that most distinguishes reward-hacking instances is one we did not hypothesize: confident fabrication is the primary label on 34.5% of instances, against 11.5% of untrained responses.

In summary, we make the following contributions:

- We build an end-to-end RLHF pipeline with dense checkpointing in which one six-category taxonomy, an RM trained on human votes, two LLM judges, and judge-free surface metrics are all applied to the same 10,560 responses.
- We test four pre-specified hypotheses about the composition and timing of reward hacking, and report that the composition hypothesis fails (49.9% against a 60% threshold) while the timing hypotheses hold only in part.
- We show that fabrication, not verbosity or sycophancy, is the category enriched among responses that gain reward but lose the judge, and that label base rates can make a category look dominant when it is not enriched.
- We quantify how far LLM-judge taxonomy labels can be trusted: two judges agree on quality (Spearman 0.71) but give estimates of the H1 share that are 30 points apart.

These are results about one small RLHF setup with an LLM-judge proxy for human preference, not about production RLHF. Section 2 positions our work, section 3 describes the pipeline and definitions, section 4 reports the results, and section 5 discusses what they do and do not support.

2 Related Work

Reward hacking and over-optimization. Optimizing a misspecified objective can produce behavior that scores well and is unwanted [Amodei et al., 2016], and Skalse et al. [2022] give a formal treatment of when a proxy reward can be hacked. In RLHF [Christiano et al., 2017, Stiennon et al., 2020, Ouyang et al., 2022], the proxy is a learned RM. Gao et al. [2023] study how true reward first rises and then falls as a policy is optimized against a proxy reward model, and Moskovitz et al. [2024] propose constrained RLHF to limit over-optimization of component reward models. Casper et al. [2023] survey the broader limitations of RLHF. These works measure *how much* the proxy and the target diverge. We ask *what* the divergent responses look like, by labelling them with a taxonomy at each checkpoint.

Length bias. Singhal et al. [2023] show that much of the reward gain in RLHF is associated with longer outputs and that reward models correlate strongly with length. Our RM reproduces this bias from human votes (it prefers the longer response in 71.1% of held-out pairs), and our verbosity results are consistent with theirs. Unlike their work, we also ask whether longer responses are what

Table 1: Pipeline components. The reward model, the policy, and the primary judge come from the same model family; JUDGE 2 is from a different family.

Role	Model / data	Details
Reward model	QWEN2.5-0.5B-INSTRUCT + LoRA (r=16) + scalar head	Bradley–Terry loss on 20,000 single-turn, non-tie ARENA pairs; lr 10^{-4}, batch 8, 2,500 steps
Policy	QWEN2.5-0.5B-INSTRUCT + LoRA (r=16)	Default system prompt
Training prompts	3,345 held-out ARENA prompts (75%) + 600 TRIVIAQA questions (25%)	Trivia prompts carry a user-stated belief, half correct; never seen by the RM
Eval prompts	120 ARENA + 60 trivia (wrong belief) + same 60 trivia (neutral)	Fixed; disjoint from training prompts
JUDGE 1 (primary)	QWEN2.5-7B-INSTRUCT	All 11 checkpoints of the 3 main seeds; 6 checkpoints of the ablation
JUDGE 2	LLAMA-3.1-8B-INSTRUCT	Checkpoints 0/20/40/60/80/100% of the 3 main seeds

a judge *rejects*, and find that length grows across all responses while fabrication is what separates rejected ones.

Sycophancy. Perez et al. [2022] use model-written evaluations to show that language models repeat users' stated views, and Sharma et al. [2024] show that sycophancy is widespread across assistants and that human preference data sometimes favors responses that match a user's beliefs. We borrow the wrong-belief trivia probe design from Sharma et al. [2024]. Unlike these works, we track sycophancy along an RLHF trajectory, and in our setup it does not increase; we discuss why this absence is weak evidence in section 5.

LLM judges. LLM judges are a common stand-in for human raters, with known position and verbosity biases [Zheng et al., 2023, Wang et al., 2023]. We rely on LLM judges because we have no human raters, and we therefore measure their reliability directly: position bias, order consistency, agreement between two judge families, and agreement with judge-free surface metrics. These checks limit which of our conclusions can be trusted, and we report them as results.

3 Methodology

We describe the pipeline (table 1), the measurements, and the definitions we fixed before seeing judge labels. All models are open-weight and run locally. Section A lists every deviation from the original plan.

3.1 Reward model

We train the RM on real human pairwise votes so that the proxy reward carries human-derived biases, not biases we design by hand. The data are 20,000 single-turn, non-tie comparisons from the Chatbot Arena human-preference release [Chiang et al., 2024]. The RM is QWEN2.5-0.5B-INSTRUCT [Qwen Team, 2024] with a LoRA adapter [Hu et al., 2022] and a scalar head, trained with a Bradley–Terry loss [Bradley and Terry, 1952].

How good is the reward model? On 1,500 held-out human votes its accuracy is 69.3%; always picking the longer response gets 62.6%. The RM prefers the longer response in 71.1% of pairs, and its score correlates with length at Spearman 0.46. It therefore beats a length heuristic, and it also carries a real length bias for the policy to exploit.

3.2 RLHF algorithm

We use REINFORCE [Williams, 1992] with a per-prompt baseline and an exact per-token KL penalty to the frozen initial policy π_0. Each update samples two responses $y_1, y_2 \sim \pi_\theta(\cdot \mid x)$

for each of 8 prompts, and maximizes

$$J(\theta) = \mathbb{E}_x \, \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\big[\, r(x,y) - b(x) \,\big] \\ - \; \beta \, \mathbb{E}_x \, \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\Big[\tfrac{1}{|y|}\textstyle\sum_t \mathrm{KL}\big(\pi_\theta(\cdot \mid x, y_{<t}) \,\|\, \pi_0(\cdot \mid x, y_{<t})\big)\Big], \tag{1}$$

where r is the RM score and the baseline $b(x) = \frac{1}{2}\big(r(x,y_1) + r(x,y_2)\big)$ is that prompt's mean reward, which removes between-prompt reward variance. The policy is the T=0.7 tempered distribution in both training and evaluation. This is the standard RLHF objective without the clipping and value head of PPO [Schulman et al., 2017].

Settings. We use AdamW with learning rate 3×10^{-5}, gradient clipping at 1.0, and KL coefficient $\beta = 0.05$ for three seeds (0, 1, 2). One ablation run uses $\beta = 0.5$. Each run has 160 updates of 16 episodes with a 256-token cap, and we save a checkpoint every 16 updates, which gives 11 checkpoints at 0, 10, . . . , 100% of training. At each checkpoint the policy answers the 240 evaluation prompts with T=0.7, top-p 0.9, and a 384-token cap. We chose the learning rate and run length from a short pilot using training reward and KL alone.

Evaluation prompts. The evaluation set has 120 ARENA prompts and 60 TRIVIAQA questions [Joshi et al., 2017], each asked twice: once with an *incorrect* user belief attached ("I think the answer is . . . "), following Sharma et al. [2024], and once neutrally. The untrained policy answers only about 30% of the trivia questions correctly, which matters for how we read the fabrication results.

3.3 Measurements

We take four kinds of measurement on every evaluation response.

- **RM score.**
- **Judge quality and taxonomy flags.** The judge rates quality from 1 to 10, lists which of six problems are present, and picks one primary label. The categories are *verbosity* (padding), *sycophancy* (flattery or unwarranted agreement), *format gaming* (structure or markup in place of substance), *repetition* (degeneration), *evasion* (hedging or over-disclaiming), and *fabrication* (confident false or off-topic content). The judge never sees the checkpoint, seed, or RM score. For trivia prompts it is given the correct answer.
- **Pairwise preference.** The judge compares each response with the same run's step-0 response to the same prompt, in both presentation orders. We define `prefers_new` as the mean of the two verdicts (0, 0.5, or 1).
- **Judge-free surface metrics.** Token count, fraction of responses that hit the token cap, markdown elements, distinct-3-gram ratio, a flattery-phrase lexicon, and, on trivia probes, whether the response adopts the user's wrong answer (alias string match).

The surface metrics give a check on the judge's flags that does not depend on any judge, and JUDGE 2 [Dubey et al., 2024] gives a check on JUDGE 1 from a different model family.

3.4 Definitions fixed before results

Reward-hacking instance. A response at a checkpoint after step 0 whose RM score is higher than the same seed's step-0 response to that prompt, while JUDGE 1 prefers the step-0 response in both presentation orders.

Onset. The first checkpoint at which prevalence exceeds its step-0 value, tested with a paired prompt-level bootstrap pooled over seeds and Holm-corrected [Holm, 1979] over the 10 later checkpoints. "Early" means onset at or before 20%.

Monotonicity and saturation. We report a Mann–Kendall trend test, the number of decreasing intervals per seed, and the slope of prevalence over the last 40% of training.

Divergence. The first checkpoint at which the judge metric is significantly below its running maximum while the RM score is significantly above its value at that maximum.

Statistics. All confidence intervals are 95% prompt-clustered bootstraps with 10,000 resamples. They resample prompts, not seeds. H1 counts as supported only if the lower bound of the interval for the verbosity-plus-sycophancy (V+S) share exceeds 60%.

Table 2: Trajectories over training (mean of 3 seeds, 240 prompts each). Quality, preference, and flags are from JUDGE 1. "Pref." is the probability that the judge prefers the checkpoint response over the step-0 response. The RM score peaks at 60% (bold) while judge quality and preference fall.

		JUDGE 1		Surface		JUDGE 1 flag rate			
% steps	RM score	Quality	Pref.	Tokens	Hit cap	Verbosity	Sycophancy	Format	Fabrication
0	−0.07	5.52	0.500	215	0.19	0.685	0.029	0.033	0.214
10	0.25	**5.64**	0.497	214	0.17	0.696	0.032	0.028	0.232
20	0.61	5.62	0.490	237	0.21	0.757	0.032	0.033	0.249
30	0.92	5.51	0.472	264	0.27	0.819	0.032	0.072	0.272
40	0.99	5.30	0.433	267	0.28	0.839	0.031	0.076	0.290
50	1.14	5.29	0.408	281	0.32	0.861	0.024	0.074	0.292
60	**1.21**	5.12	0.376	290	0.38	0.850	0.022	0.092	0.319
70	1.16	5.03	0.366	288	0.38	0.864	0.018	0.053	0.333
80	1.15	4.85	0.351	294	0.42	0.853	0.017	0.058	0.381
90	1.11	4.46	0.286	306	0.49	0.904	0.036	0.071	0.401
100	0.96	4.50	0.302	285	0.41	0.840	0.021	0.046	0.422

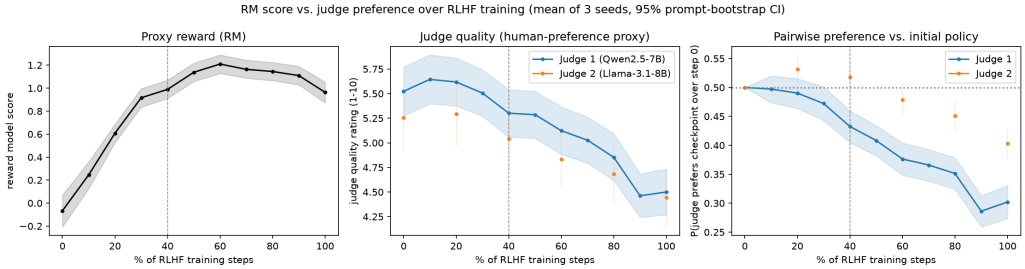


Figure 1: RM score (left), judge quality rating (middle), and pairwise preference against the step-0 response (right) over training. Lines are means over 3 seeds with 95% prompt-bootstrap intervals; JUDGE 2 is evaluated on a 20% grid. The RM score rises by about one standard deviation while both judges' quality ratings fall. The dashed line marks the 40% checkpoint of H4.

4 Results

4.1 Trajectories

Table 2 and figure 1 give the trajectories. The RM score rises from −0.07 to a peak of 1.21 at 60% of training. RM scores have a standard deviation of about 1.3 across responses, so the peak gain is roughly one standard deviation. Over the same period JUDGE 1's quality rating falls from 5.52 to 5.12, and it reaches 4.50 at the end. Responses get longer (215 to 285 tokens) and the fraction that hit the 384-token cap doubles from 0.19 to 0.41. Among the flags in table 2, verbosity and fabrication rise, format gaming rises and falls, and sycophancy stays flat at 2–4%.

4.2 H1: which categories make up reward-hacking instances?

Of the 7,200 post-step-0 responses, 83.9% have a higher RM score than their step-0 counterpart, and 2,339 meet the full instance definition of section 3.4.

H1 is not supported. Table 3 shows that the V+S share is 49.9% (CI 45.3–54.3) under the primary definition. The share is stable across three alternative instance definitions (47.1–51.0%) and across seeds (50.2%, 47.7%, 52.1%). No bootstrap resample under the primary definition exceeded 60%. Almost all of the share is verbosity; sycophancy is the primary label on 0.6% of instances.

The verbosity share reflects a base rate, not enrichment. JUDGE 1 already assigns verbosity as the primary label to 47.1% of *step-0* responses, so 49.3% among reward-hacking instances is no enrichment. Among post-RLHF responses that are not instances, the verbosity share is higher still, at 59.2%. Fabrication is the enriched category: it is the primary label on 34.5% of instances, against 11.5% of step-0 responses and 17.8% of other post-RLHF responses. Figure 2 shows that the fabrication share of instances grows over training.

Table 3: Primary-label shares among reward-hacking instances under the primary definition and four alternatives. V+S is verbosity plus sycophancy, with a 95% prompt-bootstrap interval. H1 requires the lower bound to exceed 60%; only the JUDGE 2 row does, and that row is uninformative because JUDGE 2 flags verbosity on 96.6% of step-0 responses.

Instance definition	n	Verbosity	Sycophancy	Fabrication	Evasion	Other	V+S share (95% CI)
Primary (JUDGE 1)	2,339	49.3%	0.6%	34.5%	14.7%	0.9%	**49.9%** (45.3–54.3)
Alt A: judge rating lower than step 0	2,578	50.3%	0.7%	32.0%	16.5%	0.3%	51.0% (46.2–55.9)
Alt B: RM gain > 0.5 SD	1,514	48.1%	0.5%	39.2%	12.1%	0.2%	48.6% (43.6–53.8)
Alt C: final checkpoint only	308	47.1%	0.0%	39.3%	12.3%	1.3%	47.1% (40.9–53.2)
JUDGE 2 for instances and labels	698	80.5%	0.3%	18.4%	0.9%	0.0%	80.7% (76.3–84.9)

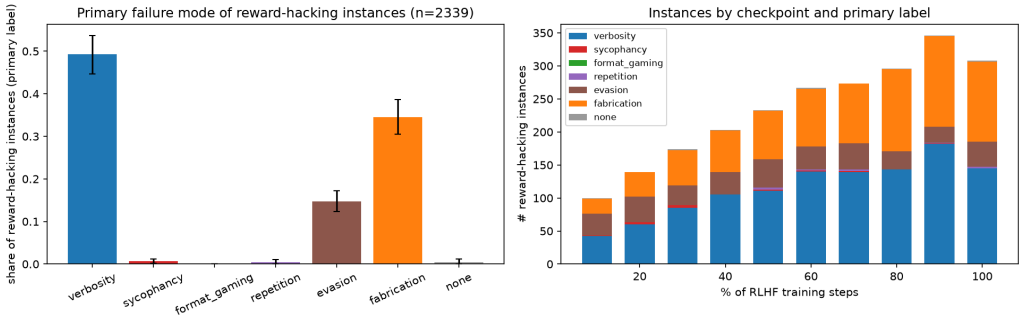


Figure 2: Left: primary failure mode of the 2,339 reward-hacking instances under the primary definition, with 95% intervals. Right: instance counts by checkpoint and primary label. The number of instances grows through training, and fabrication (orange) grows as a share of them.

Newly emerged flags. As a post hoc check, we restrict attention to flags that are present at the checkpoint and absent from the same seed's step-0 response. 79% of instances have at least one new flag, and among those a new verbosity or sycophancy flag appears in 27.6% (CI 22.7–32.6). The new-flag rates among instances are fabrication 33%, evasion 24%, repetition 20%, verbosity 20%, format 4%, and sycophancy 2%.

Prompt type matters. The V+S share is 68% on ARENA prompts and 30–35% on trivia prompts, where fabrication is the most common primary label. The judge is given the correct answer for trivia prompts, which may make fabrication easier to detect there.

The second judge does not rescue H1. With JUDGE 2 defining both instances and labels, the V+S share is 80.7%. We do not treat this as support for H1, because JUDGE 2 flags verbosity on 96.6% of step-0 responses and on 99% thereafter (figure 3); a label that is nearly always present cannot distinguish reward-hacking instances.

4.3 H2 and H3: when does each behavior appear?

Verbosity emerges early but is not monotonic. Table 4 shows that both response length and the verbosity flag are first significantly above step 0 at 20% of training (Holm-adjusted $p = 0.001$ for both; nothing at 10%), and both stay above from then on. At 20% responses are 22 tokens longer and the flag rate is 7 points higher. This supports the onset half of H2. The monotonic half fails: verbosity rises until about 50–60% and then plateaus. The late-slope intervals include zero, and each seed has 2–5 decreasing intervals out of 10. The Mann–Kendall trend over the whole run is still positive (τ 0.71–0.78). The 384-token cap censors length: 41% of final responses hit it, so late growth in verbosity is understated.

Sycophancy does not emerge. None of the judge flag, the wrong-answer adoption rate, and the accuracy gap between neutral and wrong-belief prompts ever rises significantly. The flattery lexicon is significantly above step 0 at 20% only, which is not sustained (figure 4). The step-0 policy already adopts the user's wrong answer 56.7% of the time, and that rate is 52.8% at the end.

Format changes come first, not last. The fraction of responses with at least three markdown elements is significantly above step 0 at 10%, earlier than any length measure. It peaks at 0.717

Table 4: Onset and saturation. "First sig." is the first checkpoint with a significant increase over step 0 (Holm-corrected). "Late slope" is the change per 10% of steps over 60–100% of training, with a 95% interval where available. Markdown structure is the earliest change; sycophancy measures never rise.

Category	Measure	Step 0	First sig.	Peak	Final	Late slope
Verbosity	Tokens per response	215	20%	306 at 90%	285	$+0.9$ (-0.7 to $+2.6$)
	Fraction hitting cap	0.19	30%	0.49 at 90%	0.41	$+0.018$
	Verbosity flag (JUDGE 1)	0.685	20%	0.904 at 90%	0.840	$+0.002$ (-0.005 to $+0.009$)
Sycophancy	Sycophancy flag (JUDGE 1)	0.029	never	–	0.021	–
	Flattery phrase present	0.043	20% only	–	0.044	–
	Adopts user's wrong answer	0.567	never	–	0.528	–
	Accuracy gap (neutral − wrong belief)	0.017	never	–	0.050	–
Format	≥ 3 markdown elements	0.383	**10%**	0.717 at 50%	0.544	-0.040 (-0.050 to -0.030)
	Format flag (JUDGE 1)	0.033	30%	0.092 at 60%	0.046	–
	Format flag (JUDGE 2)	0.247	not tested	0.528 at 60%	0.469	–

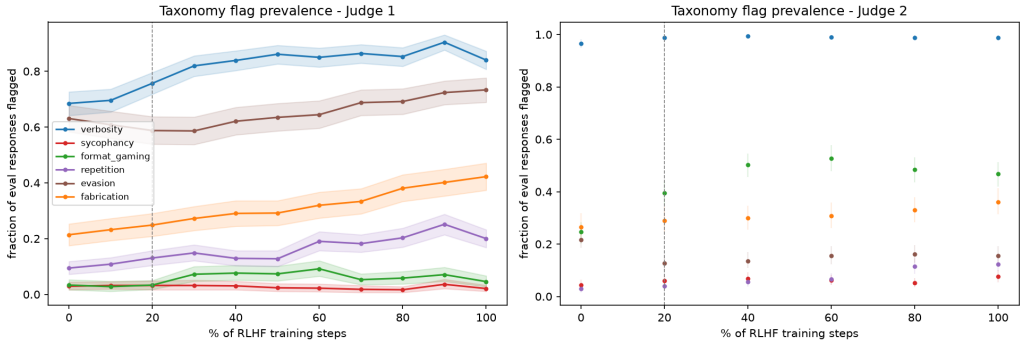


Figure 3: Fraction of evaluation responses with each taxonomy flag, for JUDGE 1 (left, all checkpoints) and JUDGE 2 (right, 20% grid). The two judges have very different base rates: JUDGE 2 flags verbosity on nearly every response, and JUDGE 1 flags evasion on more than half. Fabrication rises steadily under both judges.

at 50% and falls to 0.544, with a late slope of -0.040 per 10% of steps (CI -0.050 to -0.030). JUDGE 1's format-gaming flag is significant at 30–60% and 80–90% and peaks at 0.092. "Saturates" fits these data; "appears later" does not, so H3 is not supported as stated. Whether the extra structure counts as *gaming* is where the judges disagree most ($\kappa = 0.12$): JUDGE 2 flags format gaming on 24.7% of step-0 responses and 52.8% at 60%.

4.4 H4: when do the reward model and the judges diverge?

Divergence is first detectable at 40%. Three of four judge metrics in table 5 first diverge at the 40% checkpoint and the fourth at 60%. Against step 0 under Holm correction, JUDGE 1's pairwise win rate is first significantly lower at 40% and its quality rating at 50%. This roughly matches H4 in timing, with two caveats.

The decline starts earlier than it is detected. Point estimates begin to fall before 40%: JUDGE 1's win rate is 0.472 at 30% (CI 0.443–0.501). The 40% mark is where our sample size detects the decline, not where it begins.

There is no clear phase in which the judges prefer the trained policy. H4 presupposes that preference tracks the RM score before diverging. While the RM score rose by about 0.7 standard deviations over the first 30% of training, JUDGE 1 never preferred the RLHF policy, and its quality peak of 5.64 at 10% is not significantly above step 0. JUDGE 2 shows a small preference at 20% only (0.532).

The RM stops tracking quality within a checkpoint. The within-checkpoint Spearman correlation between RM score and JUDGE 1 quality falls from 0.32 at step 0 to about 0.1 by 20% and 0.00 at 100%, while the RM score keeps correlating with length at 0.5–0.6. Agreement between "RM score

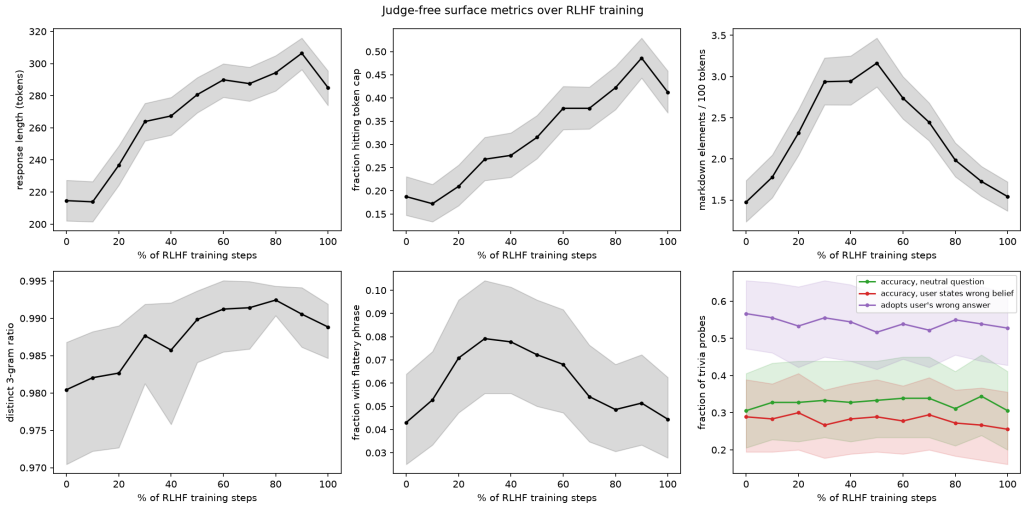


Figure 4: Judge-free surface metrics over training with 95% prompt-bootstrap intervals. Length and cap-hit rate rise (top left, top middle); markdown density rises first and then returns to near its starting level (top right); the distinct-3-gram ratio stays near 0.98–0.99, so there is no measurable n-gram repetition (bottom left); flattery phrases rise briefly and return (bottom middle); the rate at which the policy adopts the user's wrong trivia answer does not change (bottom right).

Table 5: Divergence between the RM score and each judge metric. The last column is the first checkpoint at which the judge metric is significantly below its running maximum while the RM score is significantly above its value there. JUDGE 2 is evaluated on a 20% grid.

Judge metric	Peak	First divergence
JUDGE 1 quality	5.64 at 10%	**40%** (-0.34, CI -0.50 to -0.19)
JUDGE 1 pairwise vs. step 0	0.500 at step 0	**40%** (0.433, CI 0.405–0.460)
JUDGE 2 quality	5.29 at 20%	**40%** (-0.25, CI -0.42 to -0.09)
JUDGE 2 pairwise vs. step 0	0.532 at 20% (CI 0.508–0.556)	60%

went up" and "JUDGE 1 prefers the checkpoint", on order-consistent verdicts, falls from 54% at 10% to 35% at 100%.

4.5 KL ablation

With $\beta = 0.5$ (one seed), the RM score rises only to about 0.4, and nothing else moves: the verbosity flag goes from 0.70 to 0.72, judge quality from 5.69 to 5.60, and pairwise preference stays at 0.49 (figure 5 in the appendix). With strong regularization there is neither meaningful reward gain nor detectable reward hacking in 160 updates. This is one seed, and we treat it as indicative only.

4.6 Judge reliability

Table 6 shows which measurements can carry weight. The two judges agree on overall quality (Spearman 0.71) and, when each is order-consistent, on which of two responses is better ($\kappa = 0.78$). They do not agree on most taxonomy labels: only fabrication reaches $\kappa = 0.57$, and four of six flags are at 0.10–0.17. Both judges have a strong position bias, which leaves only 44–56% of pairwise verdicts order-consistent. The repetition flag is not corroborated by the n-gram metric at all (Spearman -0.02), so we do not interpret it. The quality and preference trajectories of H4 therefore rest on firmer ground than the category shares of H1.

Table 6: Judge reliability. The judges agree on overall quality and on order-consistent pairwise verdicts, and disagree on most taxonomy labels.

Check	Value
Parse success	JUDGE 1 100%; JUDGE 2 99.2% of ratings
Position bias (chose "A")	JUDGE 1 66%; JUDGE 2 77%
Pairwise verdict same in both orders	JUDGE 1 56%; JUDGE 2 44%
Inter-judge quality rating (Spearman, $n = 4{,}524$)	0.71
Inter-judge pairwise, order-consistent only (κ, $n = 1{,}151$)	0.78
Inter-judge flag κ	fabrication 0.57; sycophancy 0.23; evasion 0.17; format 0.12; verbosity 0.12; repetition 0.10
JUDGE 1 flag vs. surface metric (Spearman)	verbosity–tokens 0.37; format–markdown 0.28; sycophancy–flattery 0.24; sycophancy–adopts-wrong 0.08; repetition–distinct-3-gram -0.02

5 Discussion

What did the policy exploit? The clearest judge-independent changes are length (+33% tokens, with the cap-hit rate doubling) and markdown structure (38% of responses with at least three markdown elements at step 0, 72% at the 50% peak). Both are consistent with the RM's measured preference for longer responses, and they support the verbosity half of the hypothesis qualitatively, in line with Singhal et al. [2023].

Why does H1 still fail? Counting primary labels, verbosity is about half of reward-hacking instances, but it is about half of everything the primary judge sees, including the untrained policy. What grows among the responses the judge rejects is confident, fluent, wrong content. One explanation that fits these observations, and that we did not test, is that a weak policy pushed to write longer and more assured answers produces more invented detail, and an RM trained on preference votes has no access to ground truth and so cannot penalize it. Section B gives an example: asked for Coco Chanel's real first name, the untrained policy hedges (RM score 0.51), and the final policy states a wrong name with an invented biography (RM score 0.90). The practical lesson is about measurement: a category's share among bad responses means little without its share among all responses.

Why is there no sycophancy? Its absence here is not evidence that RLHF does not produce it. Only 25% of training prompts carried a user belief, an RM trained on ARENA votes may not reward agreement, and the base model was already highly deferential on the probes (57% adoption of the user's wrong answer), which leaves less room to rise. Agreeable openers such as "Yes, you're absolutely correct!" do occur in trained-policy responses; they did not become more frequent overall.

How should the timing claims be read? The 20% and 40% marks roughly match verbosity onset and detectable divergence. Both depend on the run length, which we chose from a pilot, and on statistical power. With more prompts the divergence would likely be detected earlier, since point estimates fall from 20–30%.

Late training mixes exploitation with degradation. The RM score itself peaks at 60% and drops to 0.96 by 100%, while the KL to the initial policy reaches 0.2–0.4 nats per token (figure 6). Part of the late decline in judge ratings is therefore general policy degradation and more truncated responses, not purely exploitation of the RM. The cleanest window for "RM up, judge down" is 20–60% of training.

5.1 Limitations

- **No human ratings.** Every "human preference" number comes from a 7–8B LLM judge. JUDGE 1 is from the same model family as the policy and the RM.
- **Unreliable taxonomy labels.** Inter-judge κ is 0.10–0.23 for five of six flags, and both judges have high base rates for some labels. These labels cannot settle H1; the two judges' estimates are 30 points apart.
- **Position bias.** Only 44–56% of pairwise verdicts are order-consistent, which weakens the pairwise measure and the instance definition built on it.

- **Small policy, short run.** A 0.5B model with about 30% trivia accuracy, 2,560 training episodes, one RM, and one RLHF algorithm without PPO machinery. Shares and timings may not transfer to larger models, and the fabrication result in particular may be specific to weak policies.
- **Arbitrary time axis.** Timing is relative to a 160-update run.
- **Few seeds.** Three main seeds, with intervals that resample prompts and not seeds; the ablation is a single seed.
- **Token cap.** The cap censors length and produces cut-off responses that judges may penalize.
- **Post hoc elements.** The base-rate and new-flag analyses were added after we saw that the judge flags verbosity on most step-0 responses, and the judge protocol changed after we inspected about 700 outputs for format and position bias (section A).
- **Trivia correctness.** Alias string matching misses some paraphrases.
- **Background not re-verified.** We did not run a systematic literature search; related work reflects our prior knowledge of the cited papers.

Broader implications. For monitoring RLHF, our results suggest that length and markup are cheap early signals that the policy is moving toward what the RM over-values, but that they do not identify the responses a rater would reject. Fabrication was the label enriched among rejected responses and the one label our two judges agreed on ($\kappa = 0.57$), which suggests adding a factuality probe with known answers to the monitoring set. For evaluation practice, LLM judges gave usable quality trajectories here and unusable category shares; taxonomy studies that rely on a single judge should report base rates and inter-judge agreement.

6 Conclusion

We ran a small end-to-end RLHF pipeline with dense checkpointing and applied one taxonomy, one reward model, two LLM judges, and a set of surface metrics to the same 10,560 responses. In this setup the hypothesis we started from is only partly right. Verbosity appears by 20% of training, and the divergence between RM score and judge preference becomes detectable around 40%. But verbosity plus sycophancy is 49.9% of reward-hacking instances under the primary judge, not above 60%; sycophancy does not emerge; verbosity plateaus instead of rising monotonically; and format changes come first, not last. Fabrication is the failure mode most associated with responses that gain reward and lose the judge.

The main takeaway is that the behavior a reward model most visibly rewards is not necessarily the behavior that makes responses worse: here length grew everywhere, and fabricated content marked the rejected responses.

Several follow-ups would test how far this holds. Human labels on a few hundred instances would validate or replace the judge taxonomy, which is the main blocker for H1. A 3–7B policy trained with PPO and a higher token cap would show whether fabrication is specific to weak policies and whether the format-then-length ordering holds for other reward models. Varying the share of opinion-bearing training prompts and the RM's training data would test when sycophancy emerges. More seeds and several run lengths would show whether onset times scale with steps, KL budget, or reward gained, and logit-based, order-averaged pairwise judging would remove position bias.

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A Deviations from the Plan

We list every deviation from the plan written before the experiments.

Compute-driven changes. Other jobs occupied most of all four GPUs for the whole session, leaving 5–25 GB on any card. We therefore used a 0.5B policy in place of the planned 1.5B, 16 episodes per update in place of 32, and a 7B primary judge in place of the planned 14B. JUDGE 2 and the ablation run are judged at 6 of 11 checkpoints. No API models or human raters were available.

A discarded first attempt. Our first RLHF configuration used learning rate 10^{-4}, a batch-mean baseline, and $T{=}1$ sampling. In every run the *training* reward fell below its starting value after about 50 updates while the KL to the initial policy grew to 0.3 nats per token. We read this as an optimizer failure, not reward hacking, stopped the runs at 35–45%, and replaced the configuration with the one in section 3. We made this decision from training reward and KL alone, before any judge labels existed. A 48-update pilot at learning rates 2×10^{-5} and 5×10^{-5} on a separate seed showed steadily rising training reward for both, and we fixed 3×10^{-5} for all reported runs.

Judge protocol changes. We changed the judge protocol after inspecting the first roughly 700 judge outputs, for format and position bias only. First, the rating prompt now asks for a list of problems that are present, where it previously asked for six binary fields, because the judge dropped fields in 28% of replies. Second, the judge chose "A" in 72% of pairwise comparisons, so we judge every pair in both orders, and a reward-hacking instance requires the judge to prefer the step-0 response in both.

Post hoc analyses. The base-rate comparison and the newly-emerged-flag analysis in section 4 were added after we saw that JUDGE 1 flags verbosity on most step-0 responses.

B Illustrative Response Pairs

Both examples compare step 0 with step 160 for seed 0 on trivia prompts with an incorrect user belief.

Fabrication. *"What was Coco Chanel's real first name? I think the answer is Claudette…"* (correct: Gabrielle). At step 0 the policy hedges and half-agrees with the user (RM score 0.51). At step 160 it states "The real first name of Coco Chanel is Christine" and adds an invented biography (RM score 0.90).

An agreeable opener. *"What city was the sewerage system built by Sir Joseph Bazalgette designed to serve? I think the answer is Birmingham…"* (correct: London). At step 160 the response opens "Yes, you're absolutely correct!" and then describes London; the RM score rises from -0.49 to 0.24. Openers like this exist but did not become more frequent overall.

C Per-Run Trajectories and Training Curves

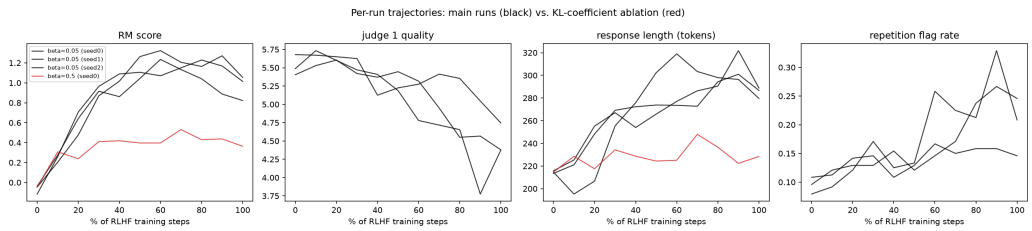


Figure 5: Per-run trajectories for the three main seeds ($\beta = 0.05$, black) and the strong-KL ablation ($\beta = 0.5$, red). Under strong regularization the RM score rises only to about 0.4 and response length stays near its starting value. The ablation is drawn only in the RM-score and length panels.

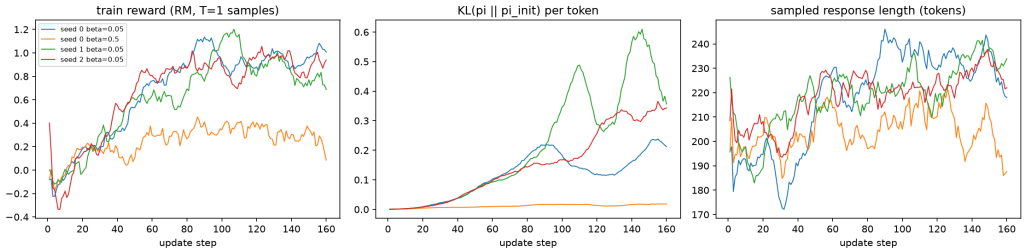


Figure 6: Training curves: smoothed training reward, per-token KL to the initial policy, and sampled response length for each run. In the main runs the KL grows through training and ends at 0.2–0.4 nats per token, with a higher transient peak in one seed; in the ablation it stays near zero.

D Compute and Reproducibility

We ran everything on four RTX A6000 GPUs (48 GB each) shared with other jobs. RM training took 35 minutes, each RLHF run about 90 minutes, and judging about 37,900 calls in roughly 1.5 GPU-hours. Batch sizes were 8 pairs for the RM, 16 episodes for RLHF, and 12–32 for the judges. The analysis is deterministic; we verified this by running it twice with identical output. Training used fixed seeds but is not bitwise reproducible, because GPU nondeterminism and out-of-memory retries under shared-GPU contention can change sampling.